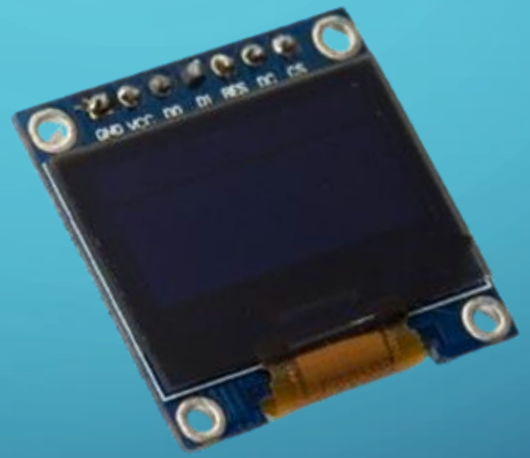
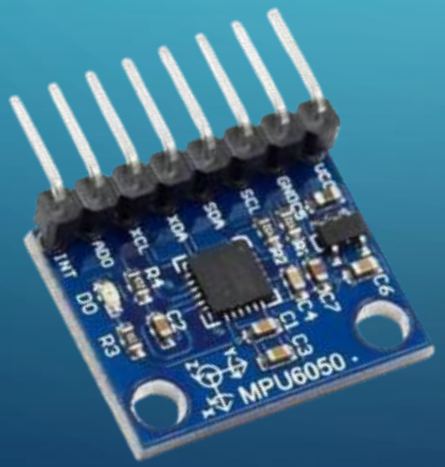
STM32 CONNECTIVITY: BLOCK BREAKER

IEEE PENN STATE - PROJECTS COMMITTEE

SP26





Attendance Form

QUICK ANNOUNCEMENT!

American Society of Agricultural and Biological Engineers (ASABE) is holding the **Robotics Student Design Competition** every year during their Annual International Meeting. This year, the Robotics Competition will be held at Indianapolis, Indiana, from **July 12, 2026 to July 14, 2026**. Please review the following:

https://asabe.org/event-detail/2026-annual-international-meeting

Scan me for more info!

ANNOUNCEMENT CONT.

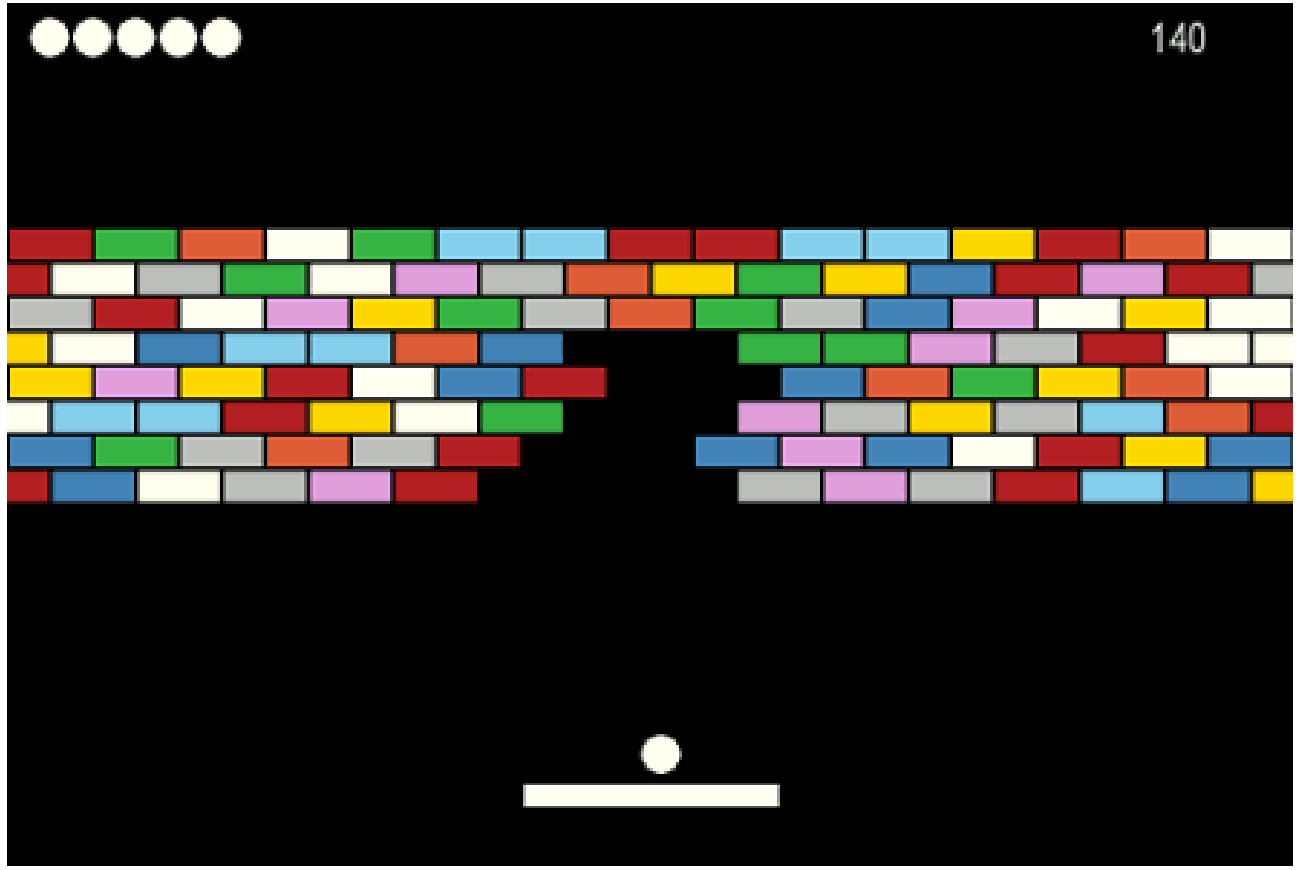
At Penn State, we are currently forming a **Beginner** Team and an **Advanced** Team to compete at the 2026 ASABE Robotics Design Competition.
If you are interested, please contact one of the faculty advisors listed below:

Student team member qualifications:

1. interested in any of robotics, programming, electrical and mechanical hardware design, Agricultural and Biological Engineering, desire to learn, etc.
2. undergraduate (freshman to senior) or graduate student, team player
3. time availability -
   * 5hr/week
   * available to travel and attend the competition meeting on **July 11, 2026 to July 14, 2026**
4. register 1 or more credits of CMPEN/BE 296, 496, or 596 for this summer

Prof. Kyusun Choi, Computer Science and Engineering
Email: kxc104@psu.edu
Office: 325 Leonhard Building
University Park
Office Hour:
1:00pm - 2:30pm Tue.
10:30am - 12:00pm Wed.

Prof. Shirin Ghatrehsamani, Agricultural and Biological Engineering
Email: spg5994@psu.edu
Office: 203 Agricultural Engineering Building
University Park



LET'S GET STARTED!

QUICK NOTE

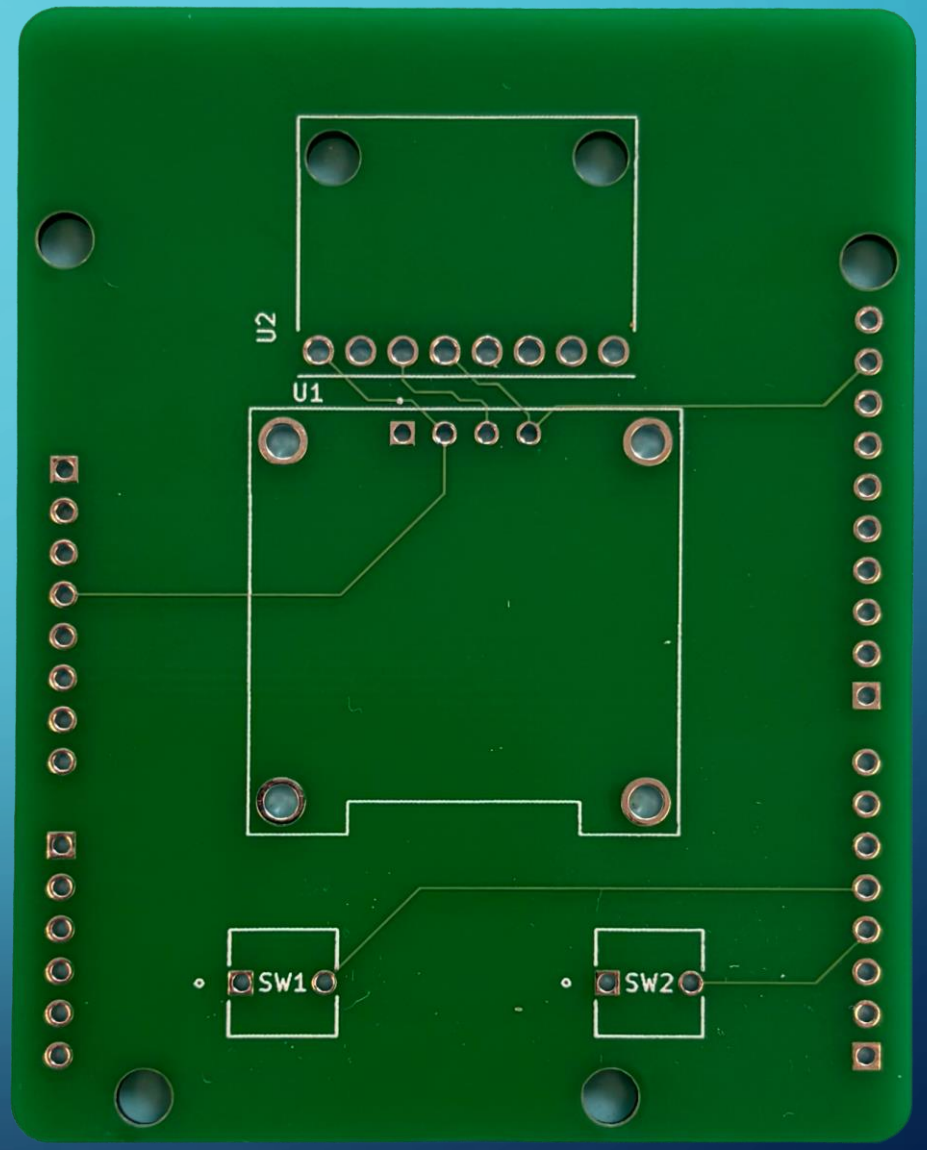
- We are assuming you have been with us through workshop 1, meaning:
  - You have **STM32CubeIDE 1.19.0** installed on your computer
  - You have seen the I/O Configurator before
  - Familiarity with your peripherals, namely the OLED screen and accelerometer

Link to Slides / GitHub

Part 1

WORKSHOP OVERVIEW

Using the Custom PCB Hat provided, we'll mount an OLED display, accelerometer, and user-input buttons to play block-breaker on our STM32!



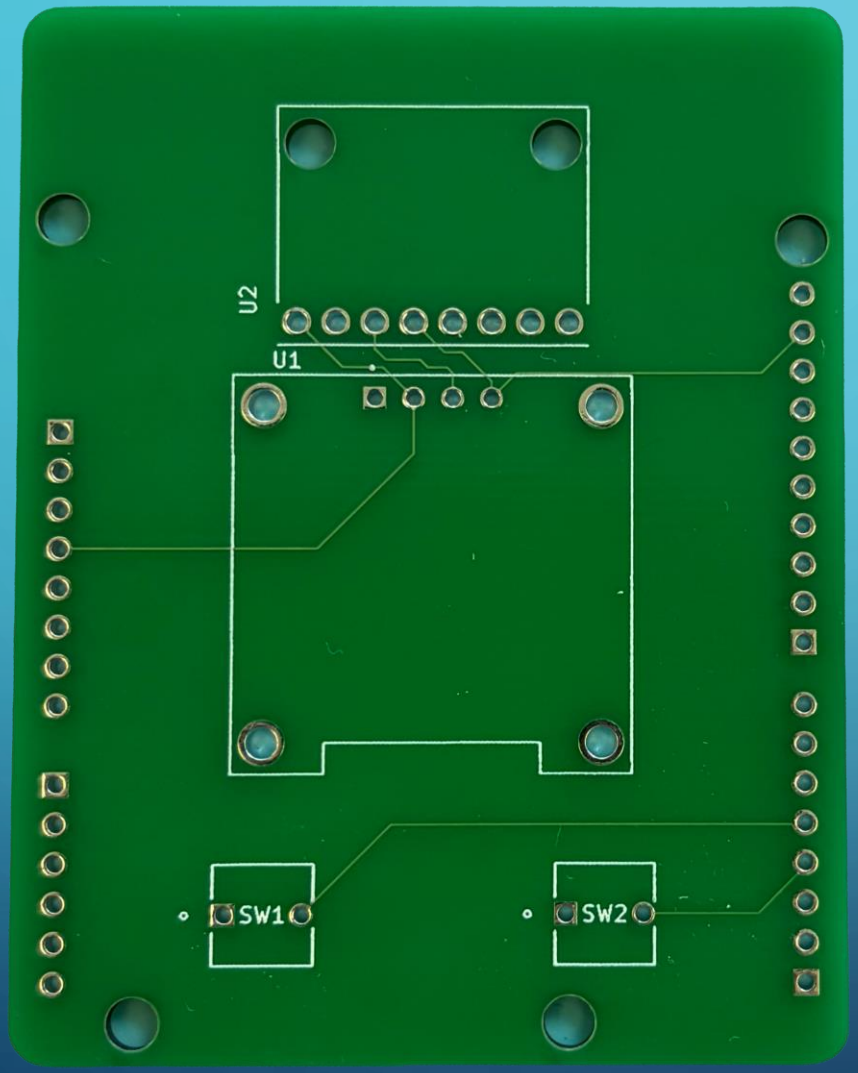
Part 2

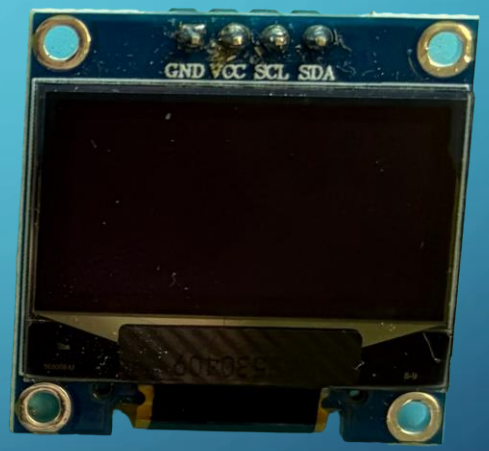
ASSEMBLY

QUICK NOTE

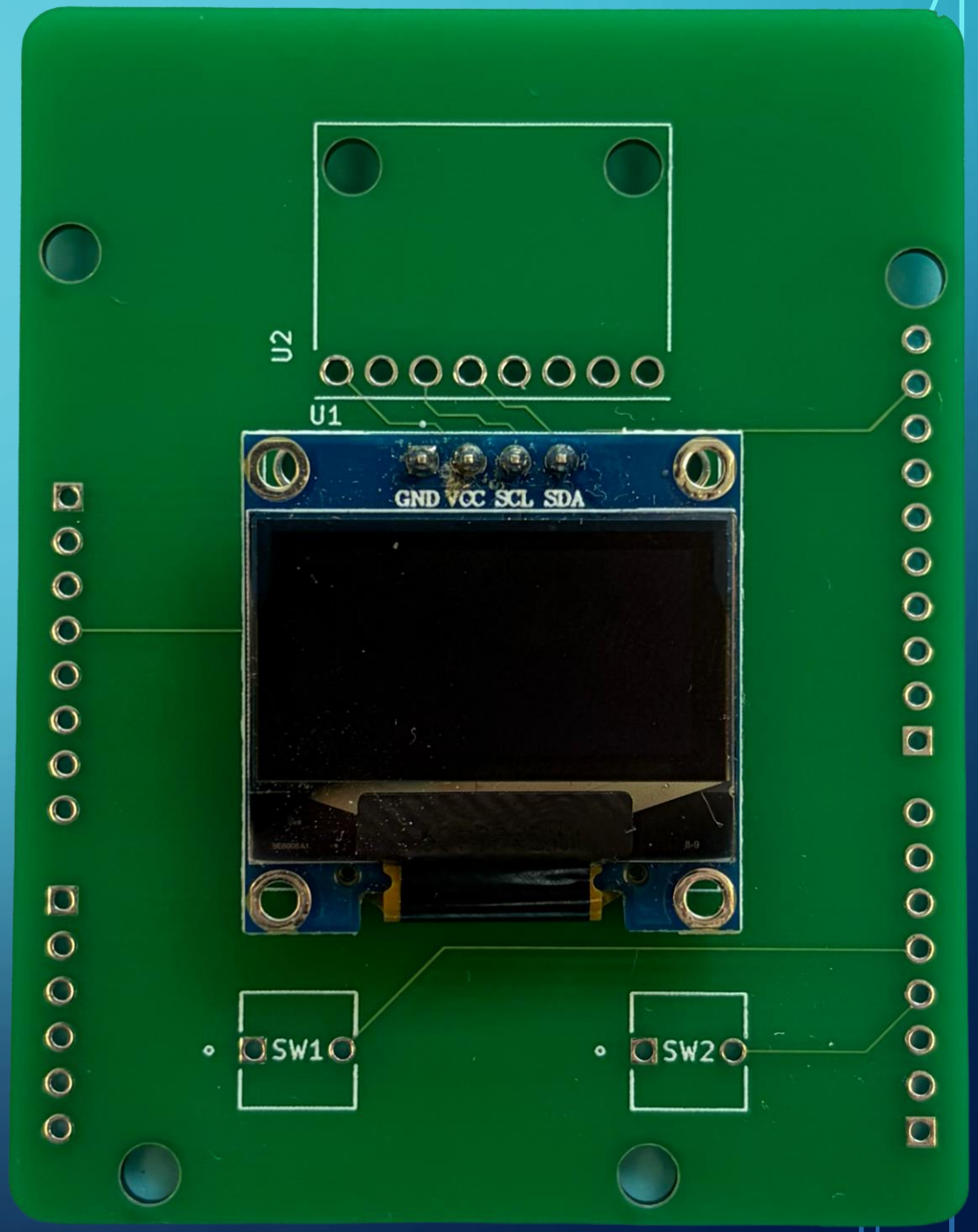
- During this process, solder carefully
- The pins and connections are small and closer together.
- If you need help, or make a mistake, don’t be afraid to call one of us over!

INSTALLING THE OLED DISPLAY

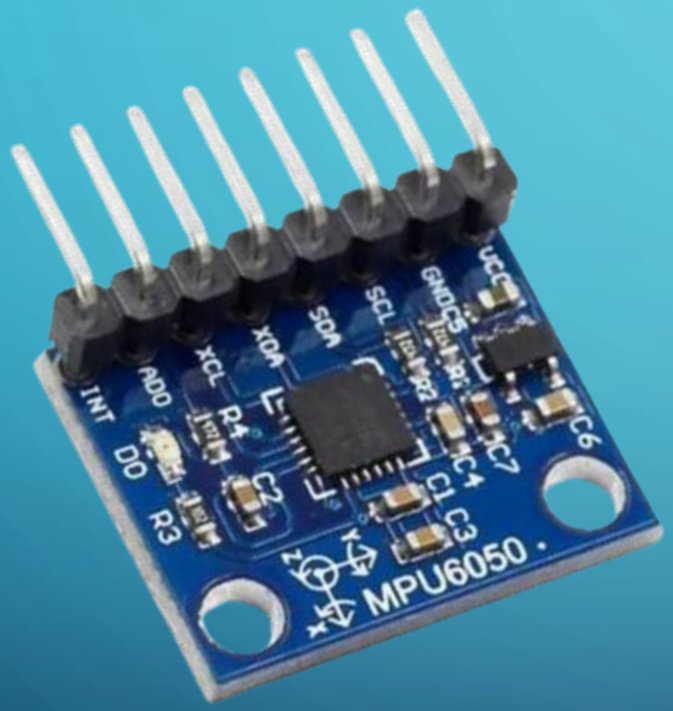


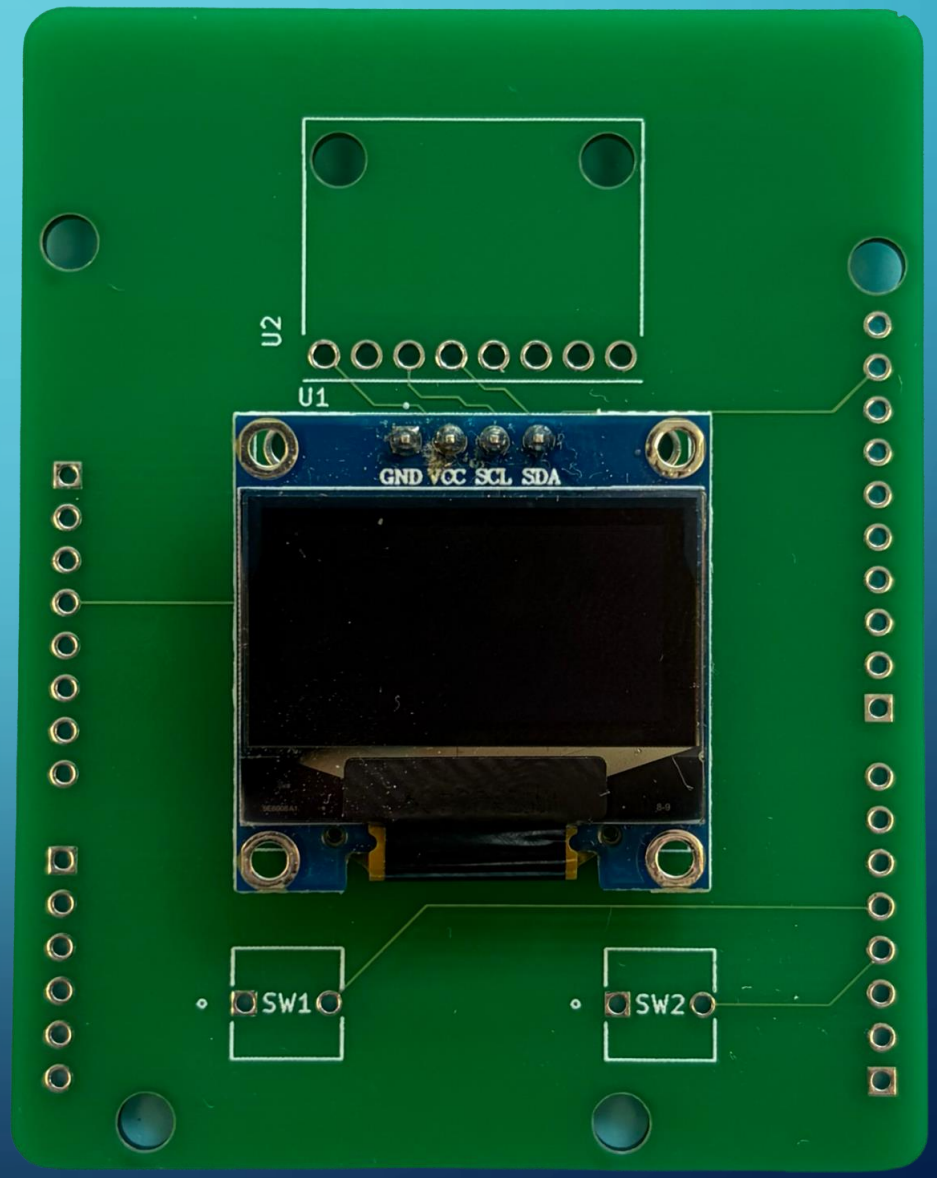


- Make sure that you have the PCB oriented correctly! (Use the side with the white outlines)
- The pins on the PCB will match those on the OLED; GND, VCC, SCL, SDA
- To solder, insert pins and carefully flip over the PCB. With the pins exposed on the backside, you can easily solder and clean up!

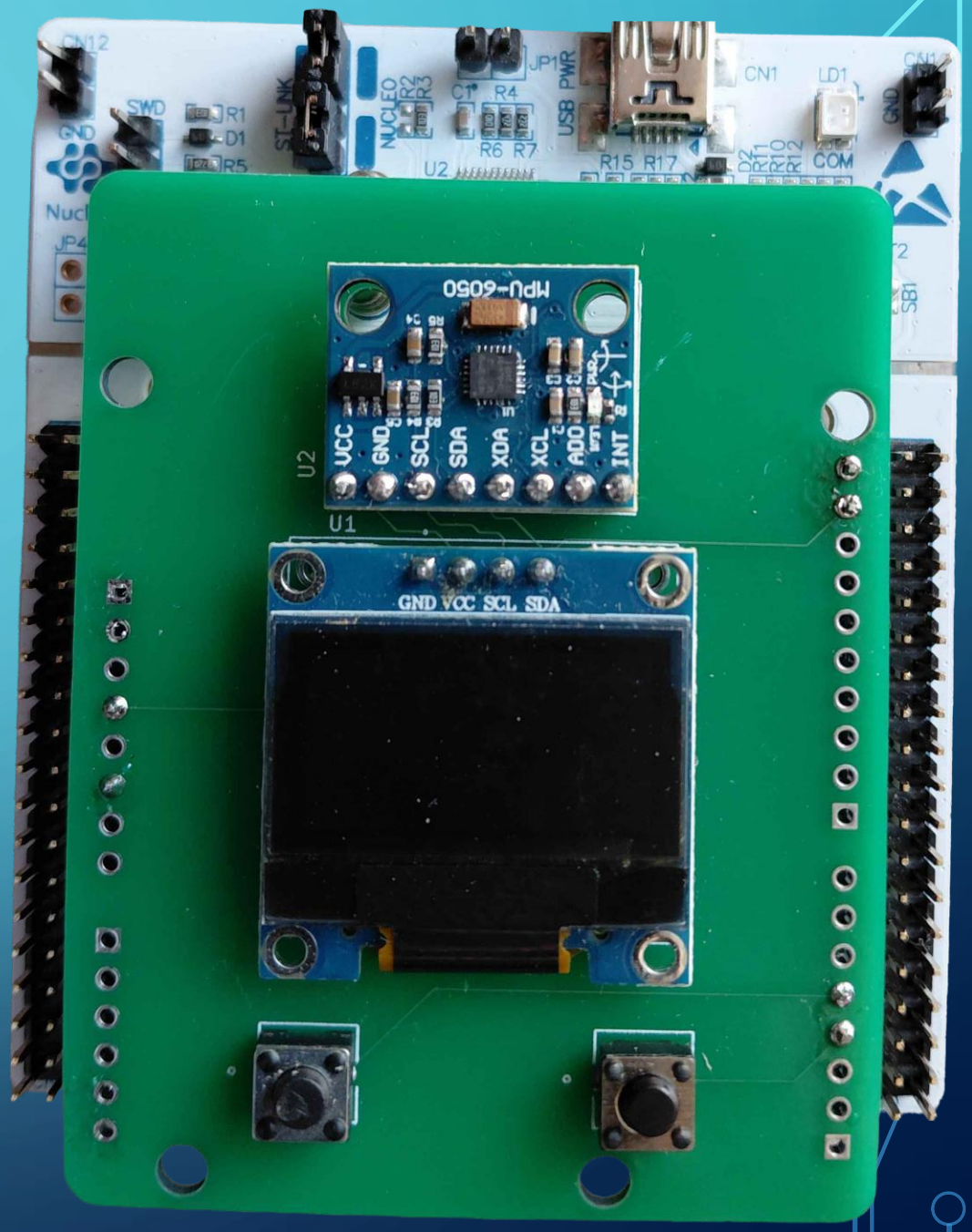


INSTALLING THE ACCELEROMETER

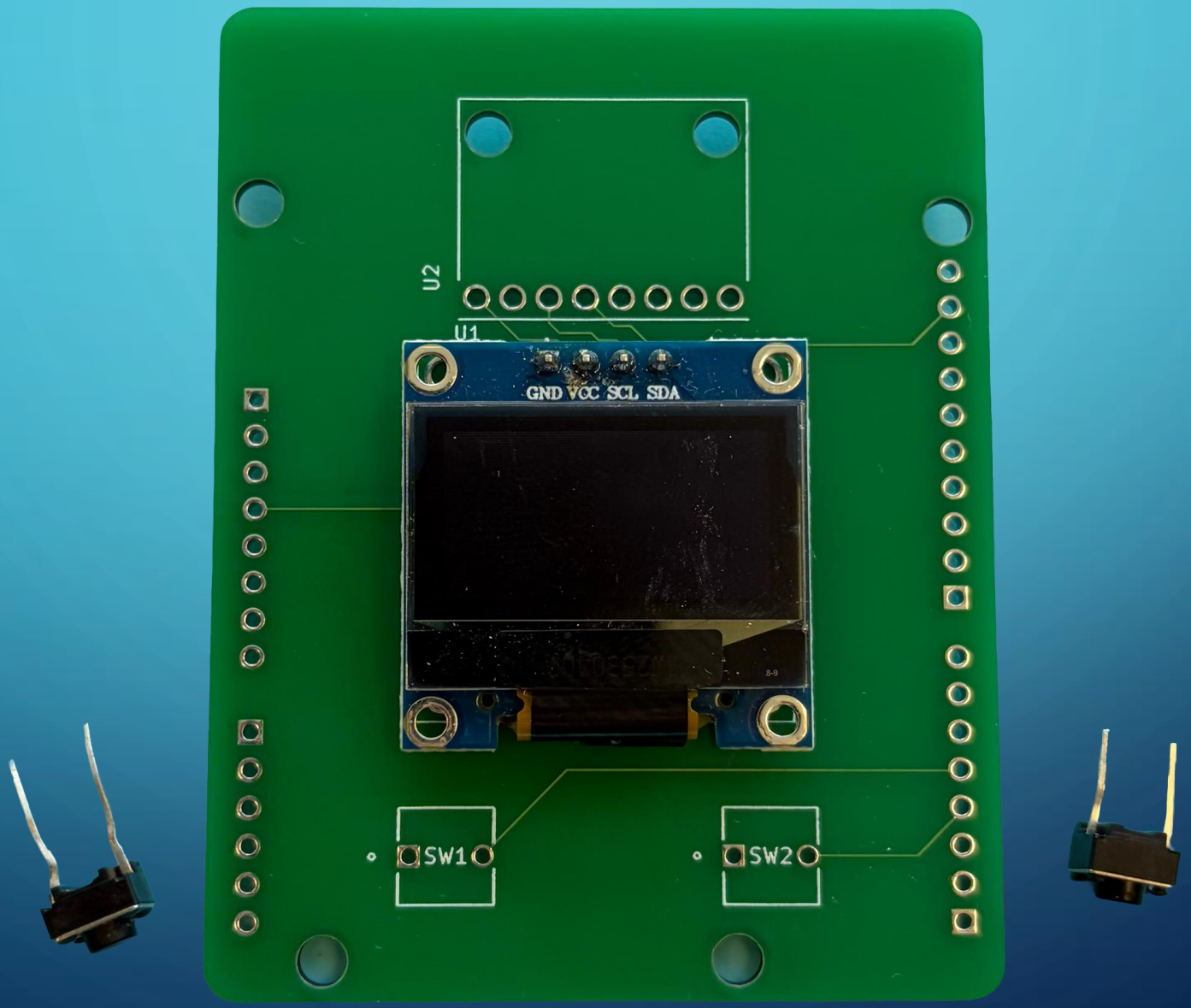




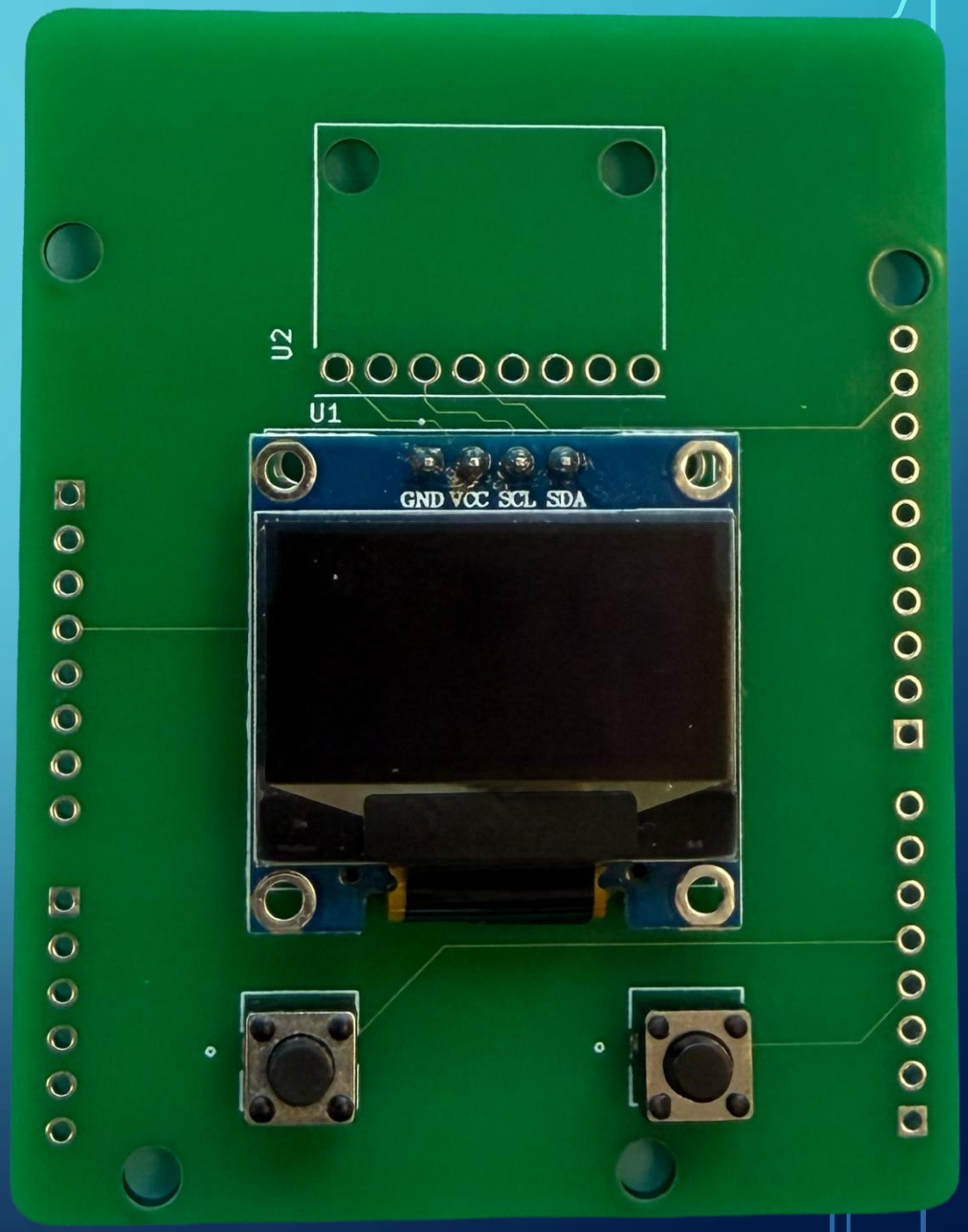
- Again, make sure that you have the PCB oriented corrected!
- This is the hardest piece to solder on, don’t rush!



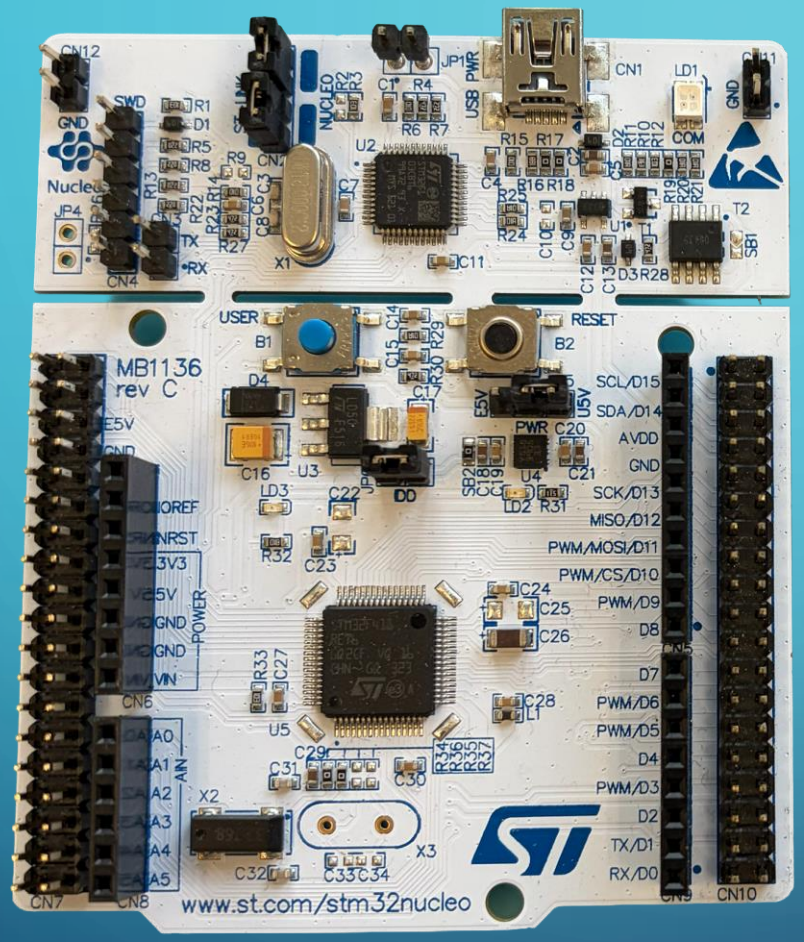
ADDING USER-INPUT BUTTONS

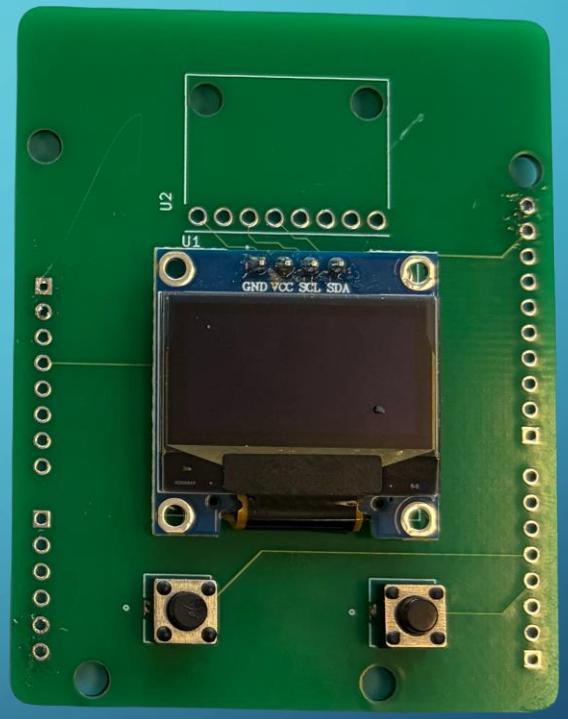


- Two push buttons will control paddle movement and are connected via GPIO pins on the STM32.
- Don’t worry about the orientation of the buttons, either direction will work.



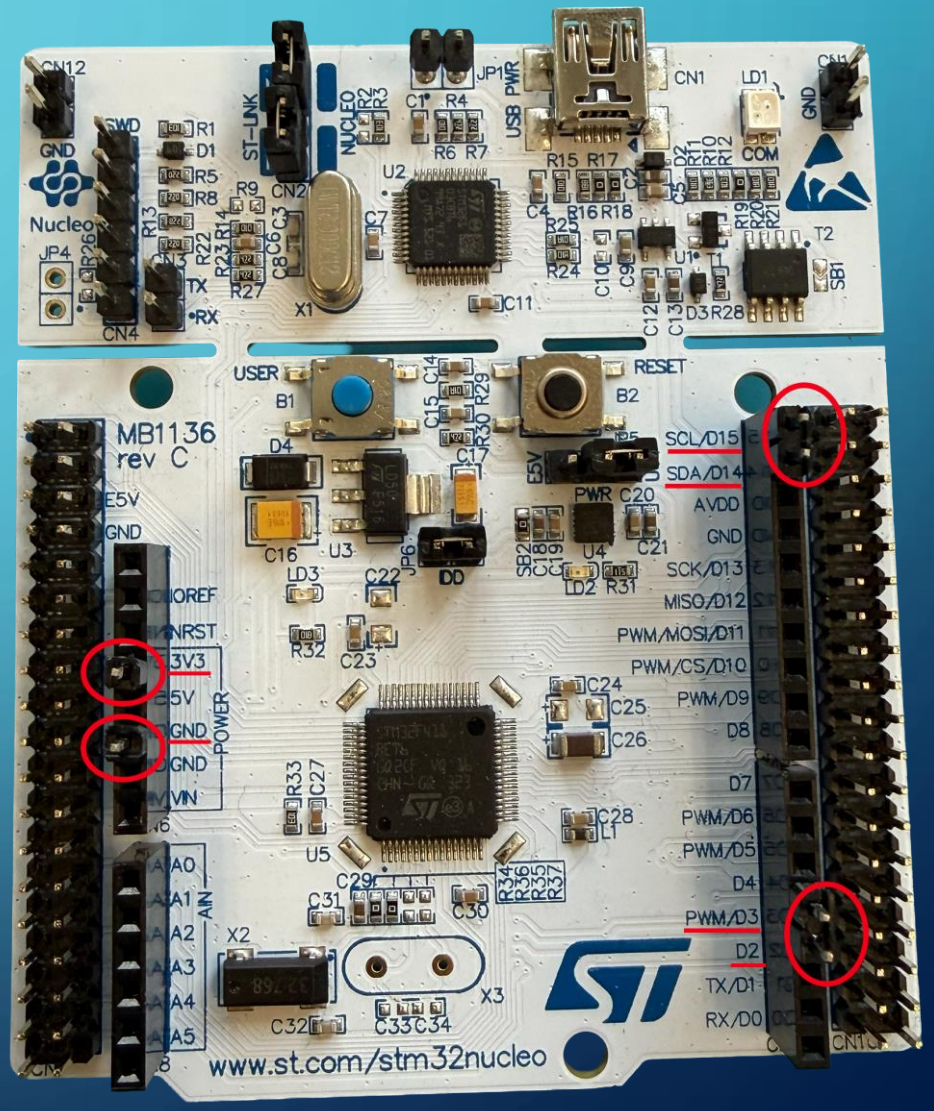
INSTALLING THE PCB HAT



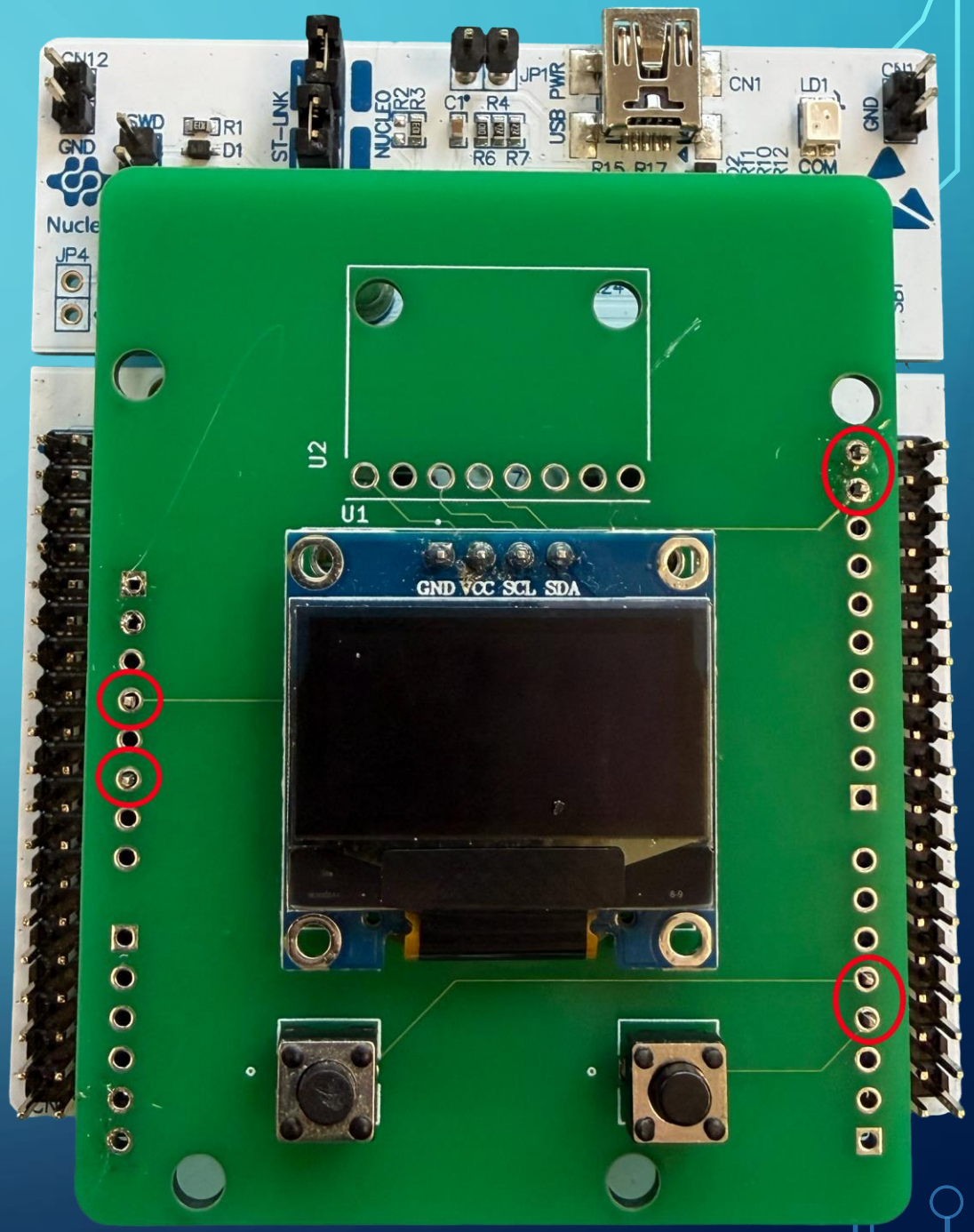


You can break your headers using the provided pliers: You'll need 2 double headers, and 2 single headers.

- Make sure you insert them into the proper pins, we'll be using:
  - 3V3 (Power)
  - GND (Ground)
  - SCL/D15, SDA/D14 (Screen Communication)
  - PWM/D3, D2 (GPIO and User-Input)



Now you can place your PCB on top of the pins and solder it in place!

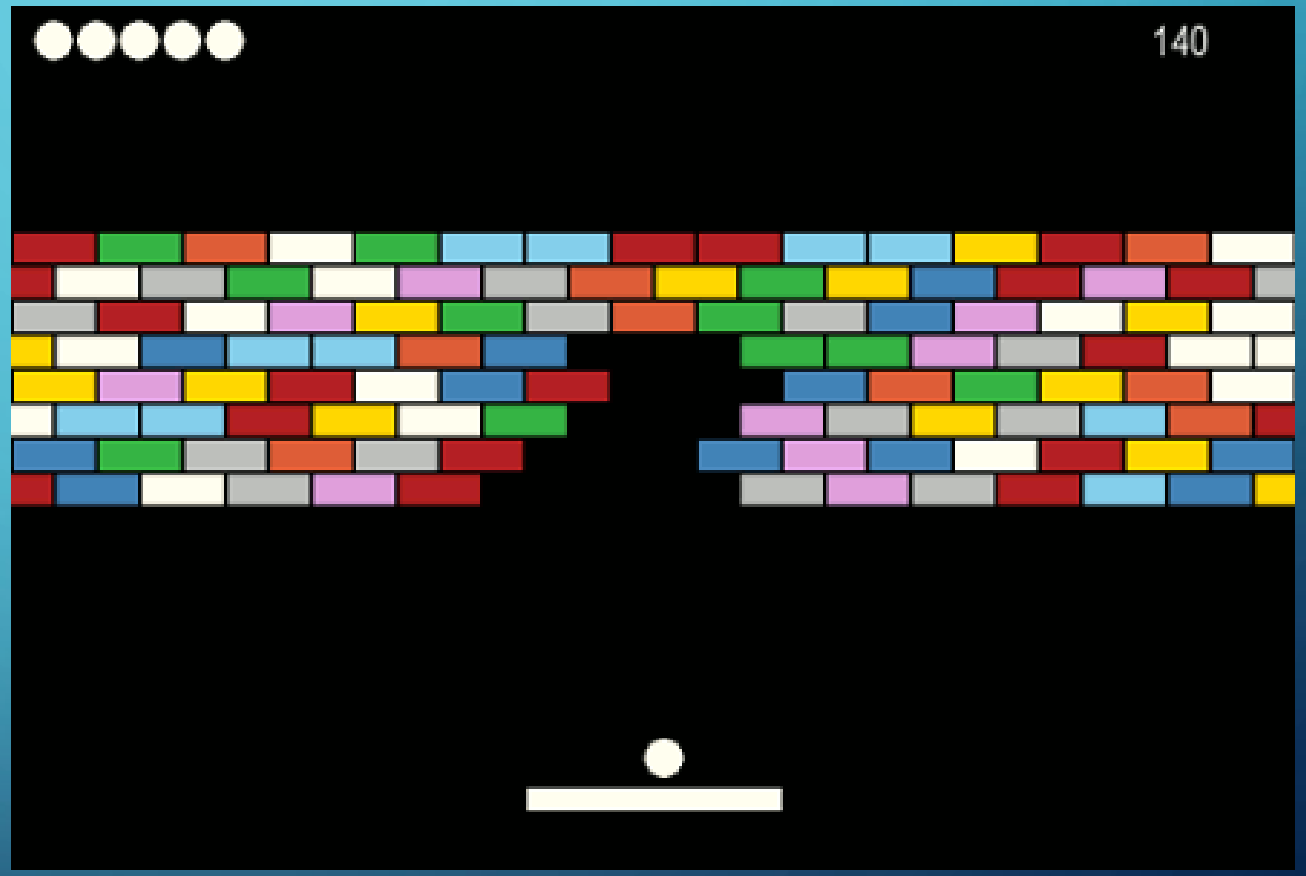


Part 3

INSTALLING BLOCK-BREAKER

Visit the GitHub for this workshop and download the code! It'll be plug and play this time around:

https://github.com/psuieee/stm32-workshop5

THANKS FOR COMING!

- It was great to see everyone's game logic in action. If you have any last-minute bugs or questions, feel free to stick around!